



* April 5, 2001
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Mattia Garatti

Curriculum Vitae

I am a doctoral researcher in Mathematics interested in Mathematical Analysis. Currently, I am carrying out my research on regularized optimal transport problem(s). I am also a private teacher, mainly for high school and university students.

Current position

Doctoral researcher in Mathematics, *Université Paris-Saclay, Orsay*, **2025 – ongoing** France.

Thesis: "On regularized Optimal Transport problems". Supervisors: Luca Nenna - Simona Rota-Nodari.

Education

Master degree in Mathematics (LM-40), *Università Cattolica del Sacro Cuore, Brescia*, **2023 – 2025**

Average: 30 cum laude. Main courses' topics: Real and Functional Analysis, Measure Theory, Differential Geometry, Mathematical Physics. Thesis: "Variational representations of optimal transport and Schrödinger problem". Supervisor: Luca Tamanini - Cosupervisor: Nicolò De Ponti.

Bachelor degree in Mathematics (L-35), *Università Cattolica del Sacro Cuore, Brescia*, **2020 – 2023**

Average: 31.33/30. Thesis: "Il Teorema di esistenza di Peano per le equazioni differenziali ordinarie", provision of a theory on ODEs in the Peano's assumptions. Supervisor: Marco Marzocchi

High School Diploma, *Liceo Scientifico 'Camillo Golgi', Breno (BS)*, **2015 – 2020** 100/100.

Publications

A PDE approach to Benamou–Brenier formula for the Schrödinger problem, **2026**
M. Garatti, L. Nenna, S. Rota Nodari, L. Tamanini, pre-print.

Talks

Entropic regularization(s) of Optimal Transport problem(s), *Università Cattolica del Sacro Cuore, Brescia*, **March, 2026**

Talk about Schrödinger problem with a focus on the Benamou–Brenier formula

Variational representations of optimal transport and Schrödinger problem, *Math PhD Days at UniTO, Turin*, **February, 2025**

Talk on my master thesis work

Research visits

Université Cote d'Azur, Nice. Visited Simona Rota-Nodari	March 16 - March 27, 2026
Università Cattolica del Sacro Cuore, Brescia. Visited Luca Tamanini	March 05 - March 06, 2026
Università Cattolica del Sacro Cuore, Brescia. Visited Luca Tamanini	January 30 - February 06, 2026
INRIA, Paris (France). Visited Luca Nenna	July 28 - August 01, 2025

Teaching

Teaching Assistant, Université Paris-Saclay, Orsay, France. Conducted problem sessions (TD) for undergraduate students (64 hours)	2026 – ongoing
Tutor, Università Cattolica del Sacro Cuore, Brescia, Italy. Conducted problem sessions for the "MathExcellence 2025" advanced mathematics program (5 hours)	June 2025
Teaching Intern, Liceo Scientifico 'G. Golgi', Breno, Italy. Delivered lectures to high school students, administered/graded tests and supported lab activities	September 2022
Tutor, Cooperativa 'Assolo', Berzo Inferiore, Italy. Provided targeted Mathematics and Physics lessons for students with SLD	August – September 2018
Teaching Intern, Fondazione Scuola Materna 'San Giuseppe', Gratacasolo, Italy. Supervised preschool children and organized early science experiments	June 2018

Work experience

Math writer, Zanichelli editore S.p.A., Bologna (BO). Teamwork with the editorial staff, autonomous creation of exercises of various difficulty on different topics to be included in future publications	2024 – 2025
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Summer schools and workshop attended

Journées SMAI-MODE , Nice.	March 2026
Conférence biennale du groupe MODE de la Société de Mathématiques Appliquées et Industrielles	
Modelling Viscoelastic Fluids , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	February 2026
Lecture series about mathematical models for viscoelastic fluids	
Calculus of Variations and applications , <i>CIRM</i> , Marseille, France.	January 2026
Workshop on the most recent advances in the field	
Geometry, duality and convexity in new OT problems , <i>IMO Paris–Saclay</i> , Orsay, France.	October 2025
Workshop on new Optimal Transport problems arising in various applications	
PGMO Days 2025 , <i>EDF Lab Paris–Saclay</i> , Palaiseau, France.	October 2025
Workshop on Optimal Transport and its interactions with Data Sciences	
Advances in Calculus of Variations and Nonlinear Analysis , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	September 2025
Workshop in honor of Marco Degiovanni's 70th birthday	
Il Problema di Plateau , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	April 2025
Lecture series about different approach to Plateau problem(s)	
Introduction to Mechanics of Swelling , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	February 2025
Lecture series about mathematical models for swelling	
Introduzione ai modelli matematici per i cristalli liquidi , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	February 2024
Lecture series about mathematical models for liquid crystals	
Interuniversity Mathematics Summer School , <i>University of Perugia</i> .	July – August 2023
Complex Analysis and PDE courses with related final exams successfully passed	
Mathematics in Ljubljana , <i>University of Ljubljana, FMF and IMFM</i> .	July 2023
Summer School with courses on Formalization of Mathematics and Introduction to Complex Dynamics	

Awards

"Smart" study award , <i>Università Cattolica del Sacro Cuore</i> , Brescia (BS).	2025
Award for 100 best students of the university	
Merit scholarship , <i>Tenaris</i> , Dalmine (BG).	2017 – 2025
Scholarships awarded on merit every year during high school and university	
Merit scholarship , <i>Consorzio del Bacino Imbrifero Montano dell'Oglio</i> , Bergamo.	2021 – 2025
Scholarships awarded on merit every year during university	
Exemption from first year university fees , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	2020
Exemption by merit	
Merit scholarship , <i>Comune di Costa Volpino</i> (BG).	2020
Scholarships awarded on merit for high school diploma	

Organization

Member of the student jury of Galileo Prize.	2024
Jury for award regarding scientific dissemination	
Member of <i>Joint Teachers – Students Commission</i>, Università Cattolica del Sacro Cuore, Brescia.	2024 – 2025
Staff member of <i>Celebration of Mind, Notte dei Ricercatori, Venite a vedere la Terra che gira</i>.	
Mathematical popularization days, Università Cattolica del Sacro Cuore	
Staff member of <i>Disfida matematica</i>.	2023 – ongoing
Mathematical team competition for high school students, local phase in Brescia	
Member of <i>Ufficio di Sezione Elettorale</i>, Comune di Costa Volpino (BG).	2021 – 2025
Scrutineer and president of polling station	
Parish volunteer, <i>Parrocchia di "San Zenone"</i>, Gratacasolo.	2015 – ongoing
Organization of summer camps and member of the local choir with budget manager functions	

Computer skills

Programming languages: Python, C, C++	Languages	Italian: Mother tongue
Mark-up languages: LaTeX, HTML		English: Fluent
		French: Fluent

Miscellanea

Worked as a background actor in “L’âge de la solitude”

I hereby authorize the use of my personal data in accordance to the GDPR 679/16 - "European regulation on the protection of personal data".

September 07th, 2026

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